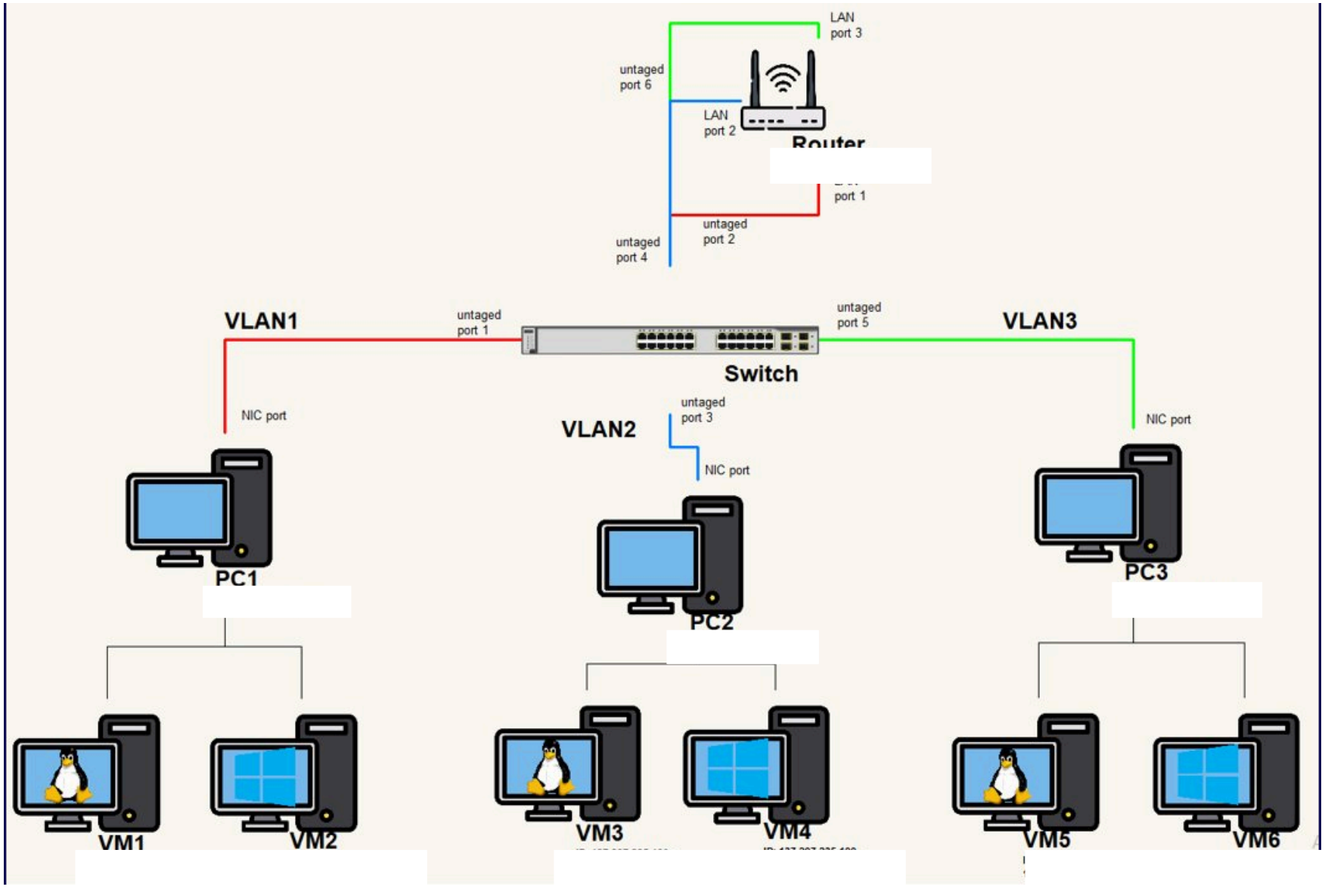
RESEARCH ENGINEERING | OCTOBER 2022 - APRIL 2023

A reproducible network-security lab

My original topology: physical hosts, virtual machines, VLAN segmentation, switching, and routing.



Penetration-testing lab setup report | Original topology figure; address labels removed for public sharing.

ORIGINAL WORK SAMPLE

Documentation that makes a setup repeatable

An excerpt from my lab setup report, explaining the environment and how the physical and virtual components fit together.

Introduction

This penetration testing environment is to be used to generate and gather data from simulated attacks on computers, and later, IoT (Internet of Things) devices deployed on protected and unprotected networks. With such a dataset, automated systems such as IDS (Intrusion Detection System) and IPS (Intrusion Prevention System) could be better trained and updated to recognize and act on various types of attacks.

This guides aims to help whoever may wish to replicate this project or take it over in the future. This guide will contain step-by-step instructions outlining each step taken to create this pen-testing environment. Since this is an ongoing project, this document will be dynamic and continuously changing, therefore, **everything in this document is subject to change**.

Before we begin, one should make sure that they have the required hardware to recreate this penetration testing environment:

- a USB drive with **at least 8GB** of space
- 3 physical computers with a **minimum of 12GB of RAM** and an **6 core CPU**
- 6 Ethernet cables
- a **managed switch** such as this one here
- a **router** such as this one here

Physical Device Specifications

Below are the hardware specifications of the PCs that we used to setup this lab environment. These specifications would be considered as the minimum requirements for such a setup.

- **Memory:** 12 GB DDR3 RAM @ 2600 MHz
- **Processor:** Intel i5-8400 6-core/6-thread CPU @ 2.8 GHz
- **Hard Disk Drive:** 1000 GB 7200 RPM SATA HDD

Project Overview

In short, this pen-testing environment will consist of a total of six virtual machines divided over three physical host devices. Each VM (Virtual Machine) will have its network card configured in a way that allows all VMs to communicate with one another as well as with each physical host (including the physical that a VM is deployed on). In order to